



**UNIVERSITY
OF LONDON**

MSc Data Science

Module: Data Visualisation

Final coursework assignment: Project

This coursework assignment is weighted at 70% of the final mark for the module.

Introduction

Following the initial exploration and practical application of Python for data preparation, manipulation, visualisation selection and evaluation, this phase challenges you to explore deeper insights in a dataset. You will be expected to demonstrate both technical fluency and critical thinking as you further develop your ability to draw more refined conclusions and produce compelling visual narratives supported by robust data handling.

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Assignment specification

Design and implement data visualisations that reveal significant insight in a given dataset of choice using suitable Python libraries and techniques. The purpose of this assessment is to demonstrate your ability to design, justify and critically evaluate a complete data visualisation workflow. Hence, produce a report as a PDF file (3000–3500 words) of a data visualisation-led investigation different from coursework assignment 1. You are required to select a relevant topic and identify one publicly and accessible dataset that are appropriate with the selected domain and apply data wrangling and visualization and draw deeper insights. Formulate five research questions that are relevant to the topic and dataset. Prepare and transform the dataset appropriately, addressing any structural formatting to ensure coherent analysis. Critically reflect on the process, discuss the limitations, assumptions, and the added value gained in the project.

IMPORTANT: beware of data found on Kaggle and other sources where the datasets are heavily processed and code for visualisation is present with multiple python codes/analysis. This will reduce your score.

Note: GenAI can be used for research with appropriate references, code debugging, data-visualisation prototypes with proper declaration. Example I acknowledge the use of [1] ChatGPT (<https://chat.openai.com/>) to [2] generate materials for background research and self-study in the drafting of this assessment. I entered the following prompts on 4 January 2026:

- [3] Write a 50 word summary about the history of Goldsmiths, University of London. Write it in an academic style. Add references and quotations from scholarly articles.

All AI contributions must be critically evaluated and cited. AI must not be used to generate full analytical reports, write entire sections of the final report, or fabricate data

Submission requirements

1. Final Report (PDF) Submission requirements

Submit your report as a PDF file and Github repository link (approx. 3,000–3,500 words)

The final report must be submitted in PDF format and prepared using a document editor. All visualisations, figures, and images generated as part of the analysis must be properly saved, clearly labelled, and embedded within the report. These should be used to illustrate key findings and support your discussion. In addition, the report must include a link to a copy of the dataset used in your analysis.

2. Jupyter Notebook (.ipynb)

You must submit a copy of the data used in your analysis, notebook (.ipynb) in Github repository

Report guidelines

Reports should include discussion of the following points.

1. Research topic and background [15%]

- Introduction/ overview of topic
- overview of key findings
- research objectives and motivation
- identify and define important concepts w.r.t research question(s) (reference academic/technical literature where appropriate)

2. Data sources [5%]

- where/how did you find them?
- how/why was the data initially collected?
- are there any ethical or legal issues?
- critically evaluate your data, is the data trustworthy, and valid for your purposes?

3. Data overview and pre-processing [10%]

- brief description of key variables
- handling of missing or erroneous data
- data summary statistics
- number of observations in the data
- summary of demographics and key variables
- use of tables or easily understandable quantities in pros

4. Analysis [50%]

- visualise key variables.
- visualise relationships between variables.
- aim for high quality explanatory visualisation that describe or tell a story about the behaviour or phenomena under investigation
- full marks will require an example of a more specialised or advanced type of visualisation, e.g., time-series, geospatial, clustering, qualitative data etc. Essentially, from topic 6-10.
- methodical data visualisation process

5. Conclusion and evaluation [10%]

- summarise key findings.
- evaluate your process and visualisations
- Challenges/limitations
- things to improve and/or pointers to future research

6. Critical Engagement with AI [10%]

- Evidence of independent thinking beyond AI output
- Correct identification of AI limitations
- Appropriate citation/acknowledgement

make sure all code runs correctly prior to submission

Penalties for exceeding the word count:

The content within the main body of text comprises the overall word count, including in-text citations, references, quotes, heading and sub-headings. The cover page, reference list and any appendices do not count towards the overall word count.

There are penalties for exceeding the specified word count.

- The maximum word limit for this coursework assignment is 3,500 words (excluding the list of references).

- You **MUST** state an accurate word count (excluding the list of references) at the end of your work. If you do not state an accurate word count your mark will be reduced by 5 marks.

- The content within the main body of text comprises the overall word count, including in-text citations, references, quotes, heading and sub-headings. The cover page, reference list and any appendices do not count towards the overall word count.

- For coursework elements and the project, there is a maximum word limit. If you exceed the word limit, we will reduce the mark you receive as follows:

Excess number of words over the word limit	Penalty applied
Up to and including 10%	5 marks deducted from original mark
More than 10% up to and including 20%	10 marks deducted from original mark
More than 20%	10 marks deducted from the original mark. The updated mark will be capped at a maximum of 40%.

Bibliography

1. <https://github.com/awesomedata/awesome-public-datasets>

2. Qin, X., Luo, Y., Tang, N. et al. Making data visualization more efficient and effective: a survey. The VLDB Journal 29, 93–117 (2020). <https://doi.org/10.1007/s00778-019-00588-3>

3. Lavanya, Addepalli, et al. "Assessing the performance of Python data visualization libraries: a review." Int. J. Computer. Eng. Res. Trends 10.1 (2023): 28-39.

4. Kelleher, Christa, and Thorsten Wagener. "Ten guidelines for effective data visualization in scientific publications." Environmental Modelling & Software 26.6 (2011): 822-827.

[END OF COURSEWORK ASSIGNMENT]